CS202: Software Tools and Techniques for CSE

Lecture 1

Shouvick Mondal

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Learning Outcomes

The students will understand the fundamentals and gain **hands-on experience** with **software tools** used by **academicians and researchers** in their day-to-day activities.

The course will prepare the students with the **necessary groundwork** and **exposure to the right mix of tools** before working with **computer systems** and undertaking **mainstream research** in domains such as analytical and experimental research.

About the Course

Instructor: **Shouvick Mondal**

TAs: Mukul Paras Potta, Vaishnav Koka, Isha Jain, Devansh Yadav

Course email (*all announcements will be made*):
cs202_stt_cse-jan-may-2025.pvtgroup@iitgn.ac.in

Google classroom (*all lab assignments will be posted*):
https://classroom.google.com/c/NzMzNDk5NDA1NjM5?cjc=3ljgqpj

Lectures: (Slot **K1**): Mon, 3:30–4:50 PM @ AB10/202
Lab sessions: (Slot **C2+D2**): Thu, 08:30–11:20 AM

- Batch 1.1 (@ AB10/104)
- Batch 1.2 (@ AB10/105)

Overall course load: {**13L** (x1)} + {**24P** (x1)}

- **Blue (lec.)**: AB10/202 (150x)
- **Red (lab.)**: AB10/104 (70x), AB10/105 (70x)

Month	Day
January	6, 13, 20, 27 9, 16, 23, 30
February	3, 10, 17, 24→ ? 6, 13, 20, 27
March	17, 24 20, 27
April	7, 14, 21 3, 17

Course Evaluation

[**A1**: 12 marks] Assess'm. I on reports/deliverables (labs cond. Jan 9, 16, 23, 30) **[Deadline: Feb 3]**

[**E1**: 20 marks] Exam I **[Mar 01–07]**

[**A2**: 12 marks] Assess'm. II on reports/deliverables (labs cond. Feb 6, 13, 20, 27) **[Deadline: Mar 10]**

[**A3**: 12 marks] Assess'm. III on reports/deliverables (labs cond. Mar 20, 27) **[Deadline: Mar 31]**

[**A4**: 12 marks] Assess'm. IV on reports/deliverables (labs cond. Apr 3, 17) **[Deadline: Apr 23]**

[**E2**: 20 marks] Exam II **[Apr 24–30]**

[**B**: 12 marks] Attendance (TAs must record **attendance in physical signature** for both lectures and lab. sessions)

All deadlines are firm, set at 23:59:59 hrs IST

Deliverables: Structured Reports + Codes

Report for assessment Ai must be submitted (<=TBA pages single column) as **`<rollno>_Ai.pdf`** *//No other extension allowed*

Codes for assessment Ai must be submitted as **`<rollno>_Ai.zip`** *//No other extension allowed*

For all submitted reports/codes, TAs will generate similarity reports from either Turnitin, moss, or other plagiarism detection tools.

Allowable similarity ranges: {reports (to be checked by TAs): <=20%, codes (to be checked by TAs): need-specific}

Deliverables: Structured Reports + Codes

Assessment criteria on structured reports (+ codes submitted)	Marks (total 12)	Description of components
Introduction, Setup, and Tools	2	Overview, objectives, environment setup, tools, and versions used.
Methodology and Execution	5	Step-by-step procedure, code snippets, outputs, screenshots, and error handling.
Results and Analysis	2	Outputs, observations, key insights, and comparisons.
Discussion and Conclusion	2	Challenges, reflections, lessons learned, and summary.
Format and Writing Style	1	Report structure, grammar, and writing style (coherent, unambiguous).
Plagiarism Compliance		**Zero marks if similarity exceeds** (reports: >20%, codes: need-specific).

Software Environment

Linux based softwares/packages are available through the following Virtual Machine (VM) image.

- Mondal, S. (2024). ***Virtual Machine for Software Engineering and Testing Group - IIT Gandhinagar***. Zenodo. https://doi.org/10.5281/zenodo.10467159.

Windows based softwares/packages are available from https://visualstudio.microsoft.com

- Download and install ***Visual Studio Community 2022***.
- *Troubleshooting*: No more support for ASP.NET Web Forms (.aspx) in Visual Studio 2022?
- *Troubleshooting*: A network-related or instance-specific error occurred while establishing a connection to SQL Server.

Course Contents

(tentative and subject to change at the sole discretion of the instructor)

Introduction to version controlling: local, centralized, distributed.

Agile development methods: Continuous integration/delivery/deployment. e.g., Git workflow, actions etc. Source code differencing algorithms in Git: myers, minimal, patience, and histogram.

Makefiles and build tools: Dependency rules, macros, suffix rules etc. build tools e.g., Gradle, Apache Maven etc.

Rapid application development strategy: introduction to event-driven programming and its structure, object based programming using windows form controls and event handling.

Data and connectivity mechanisms: Data binding in windows forms, connecting controls to data sources, displaying, and updating data.

Concept of Debugging/profiling: Visual studio debugger, GNU debugger, Linux perf – stat, trace, sampling and off-cpu profiling, hotspot UI frontend.

Learning Resources

Online:

- Continuous Integration and Delivery (***CircleCI***: https://circleci.com)

Textbook:

- Sharp, J. (2022). ***Microsoft Visual C# Step by Step***, 10th edition, Microsoft Press.
- Watson, K., Nagel, C., Pedersen, J. H., Reid, J. D., & Skinner, M. (2008). ***Beginning Microsoft Visual C# 2008***. John Wiley & Sons.
- Mark J. Price (2024). ***C# 13 and .NET 9 – Modern Cross-Platform Development Fundamentals***, 9th edition, Packt Publishing Ltd.

Reference:

- Soni, M. (2016). ***DevOps for Web Development***. Packt Publishing Ltd.
- Yusuf Sulistyo Nugroho, Hideaki Hata, and Kenichi Matsumoto. 2020. ***How different are different diff algorithms in Git? Use*** `--histogram` ***for code changes***. Empirical Softw. Engg. 25, 1 (Jan 2020), 790–823.

Introduction to version controlling: local, centralized, distributed.

We all use some software for our daily needs

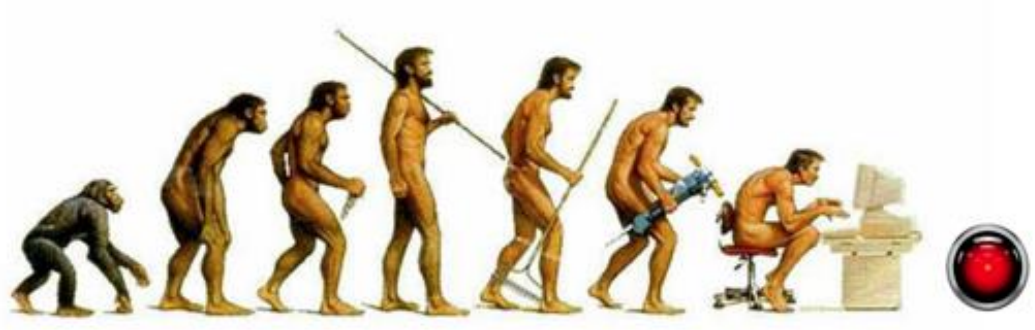
But is my software okay?
How do I know that I can trust it?

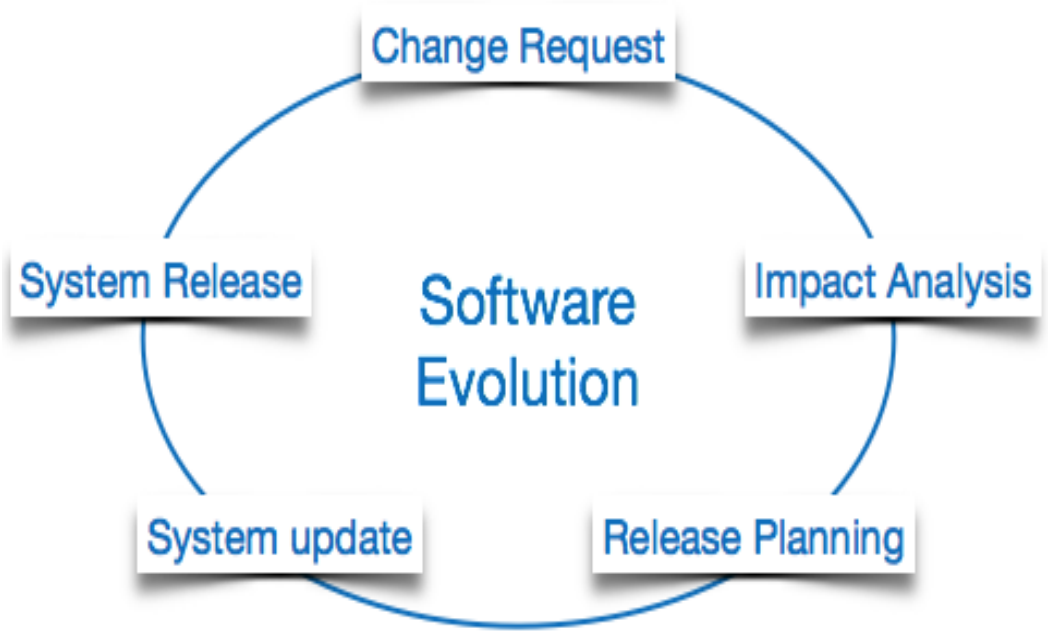
Software measurement
Testing is one way to do it...

Quality assurance: once is NOT enough!

Our needs change over time. To synchronize, software also undergoes changes. Testing them is a repetitive process…

Evolution and the process to deal with it...

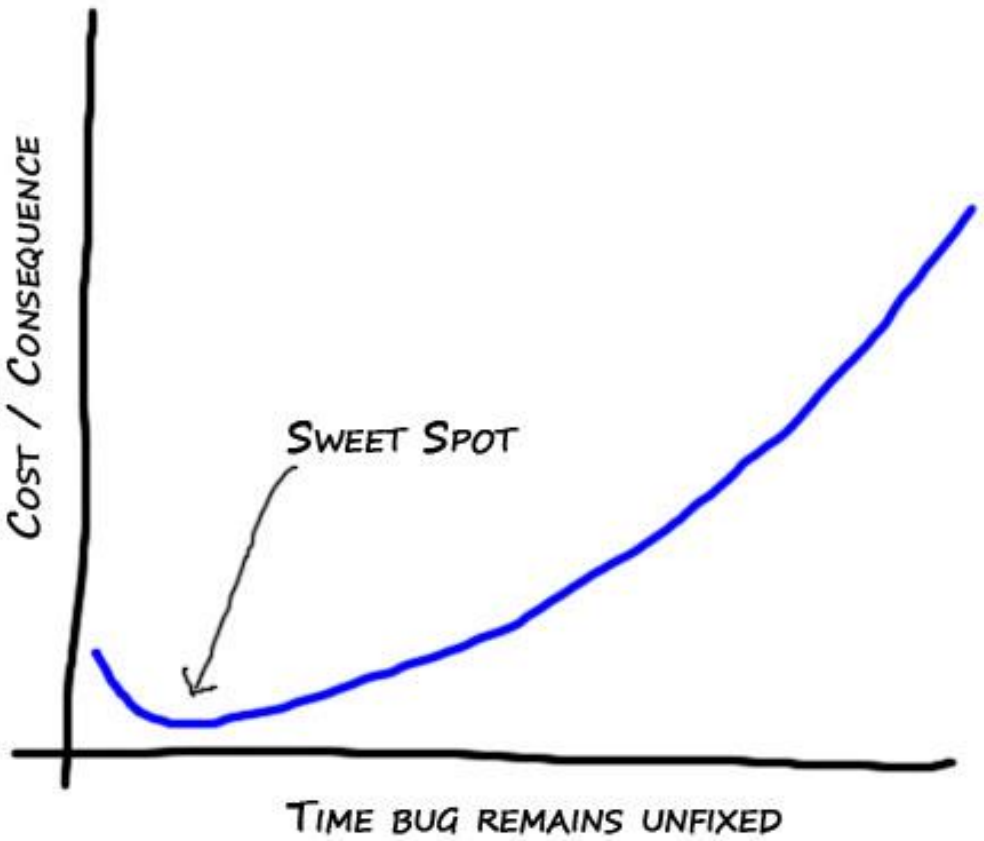


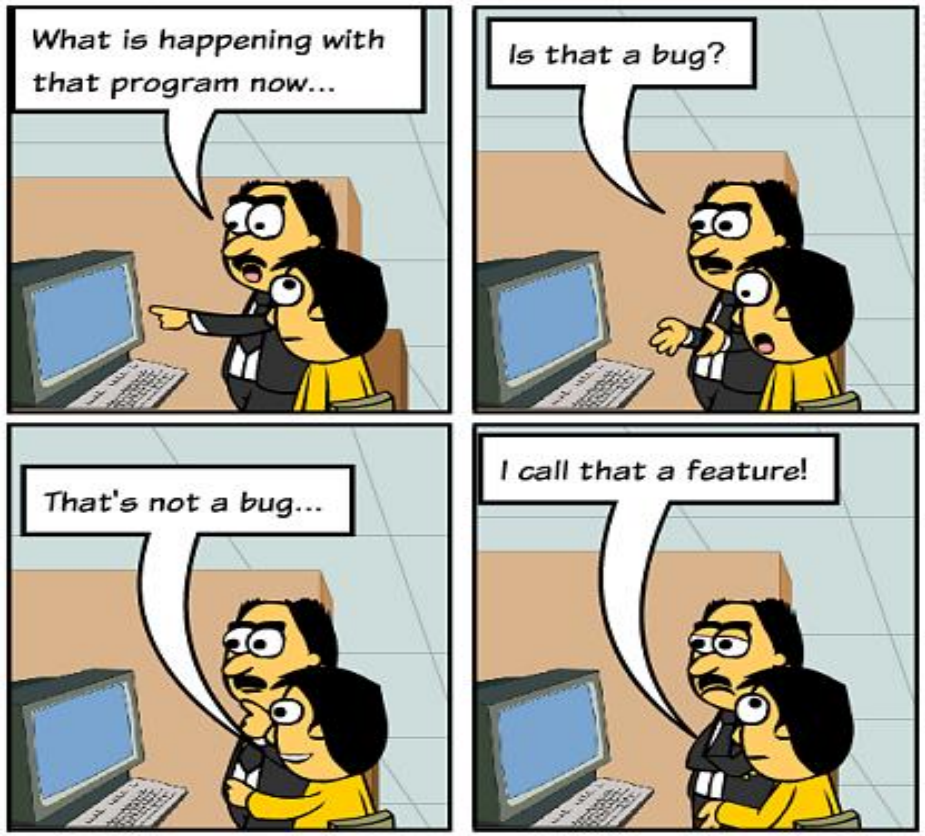


What are we testing for? BUGS

Bugs incur loss of assets...

Timing is important: the earlier the better…





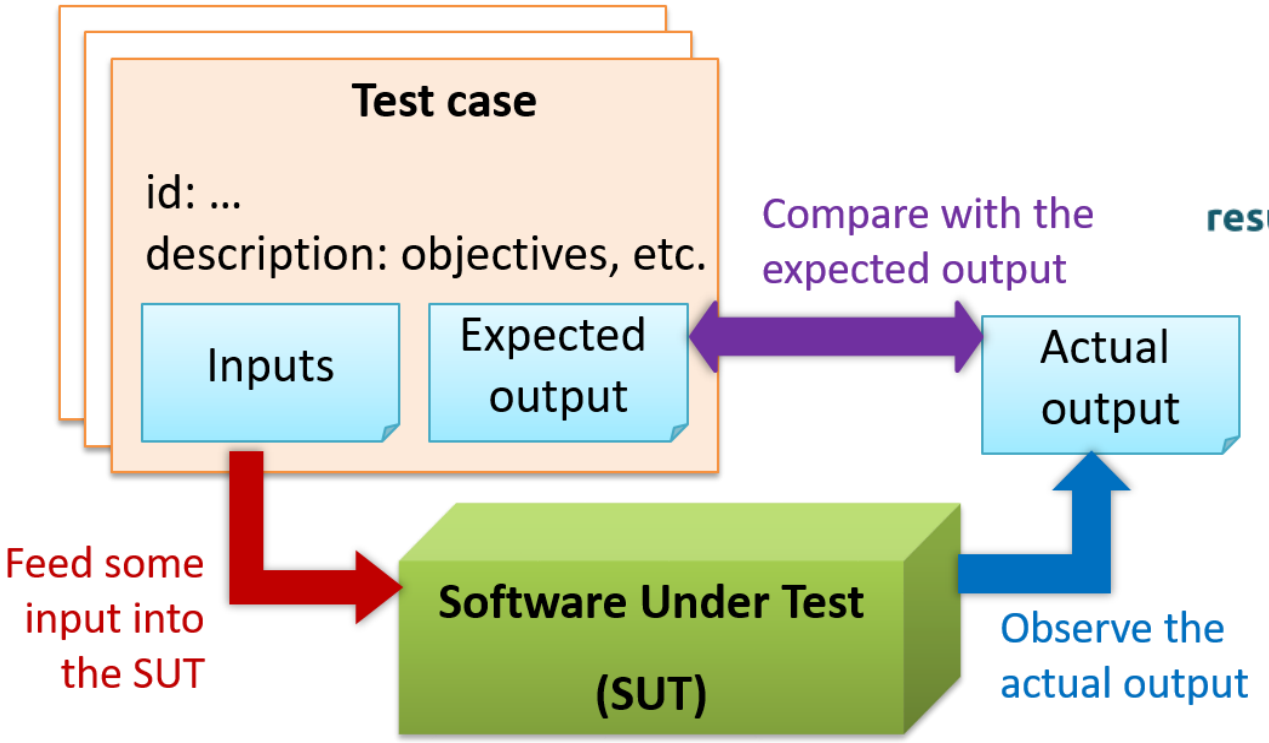
Usual software bugs

- **Functional bugs.** These are failures in the general workflow of the app caused by **improper system behavior** or enabled product features.
- **Visual bugs**. Layout and user interface **distortions** or mistakes.
- **Syntactic bugs.** The **grammar mistakes** or misspelled words and sentences used in product GUI.
- **Performance bugs**. Problematic **slowness**, **hanging**, or sluggish interface.
- **Crash bugs**. The **unexpected failure** of the program to work and function at all.

Unusual software bugs

- **Heisenbug**. The greatest trick of this bug is that it **disappears or alters its behavior when one attempts to fix it**.
- **Bohr Bug**. This unusual error **replicates itself** many times and **manifests reliably under a possibly unknown** but well-defined set of conditions.
- **Mandelbug**. Usually, it underlines **causes that are so complex** and obscure to predict. Mandelbug has **chaotic behavior**.
- *There are other species too!*

How do we test for software bugs? **Test cases**



Test case: a scenario/use case/input which **executes** the software to *observe some interesting events*.

Example of testing-and-debugging a C program (for division): *version **0** with **4** test cases*

```c
//original program
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=x/y;
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

```c
//statement of interest
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=x/y;
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

Example of testing-and-debugging a C program (for division): *version **0** with **4** test cases (using gcc 9.3.0)*

```c
//statement of interest
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=x/y;
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

```
$ ./a.out < t1
1/1 = 1.00

$ ./a.out < t2
5/4 = 1.00

$ ./a.out < t3
3/4 = 0.00

$ ./a.out < t4
15/5 = 3.00
```

Example of testing-and-debugging a C program (for division): *version **1** with **4** test cases (using gcc 9.3.0)*

```c
//statement of interest
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=(float)(x/y);
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

```
$ ./a.out < t1
1/1 = 1.00

$ ./a.out < t2
5/4 = 1.00

$ ./a.out < t3
3/4 = 0.00

$ ./a.out < t4
15/5 = 3.00
```

Example of testing-and-debugging a C program (for division): *version **2** with **4** test cases (using gcc 9.3.0)*

```c
//statement of interest
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=(float)x/y;
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

```
$ ./a.out < t1
1/1 = 1.00

$ ./a.out < t2
5/4 = 1.25

$ ./a.out < t3
3/4 = 0.75

$ ./a.out < t4
15/5 = 3.00
```

Example of testing-and-debugging a C program (for division): *version **2** with **7** test cases (using gcc 9.3.0)*

```c
//statement of interest
#include<stdio.h>
int main(void)
{
    int x,y;
    float res;
    scanf("%d %d",&x, &y);

    res=(float)x/y;
    printf("%d/%d =%.2f\n",x,y,res);

    return (0);
}
```

```
$ ./a.out < t1
...

$ ./a.out < t5
1/0 = inf

$ ./a.out < t6
0/0 = -nan

$ ./a.out < t7
0/1 = 0.00
```

Agile development methods: **Continuous** integration/delivery/deployment. e.g., Git workflow, actions etc. Source code differencing algorithms in Git: myers, minimal, patience, and histogram.

DevOps is a combined approach of Software Development (Dev) and IT Operations (Ops)

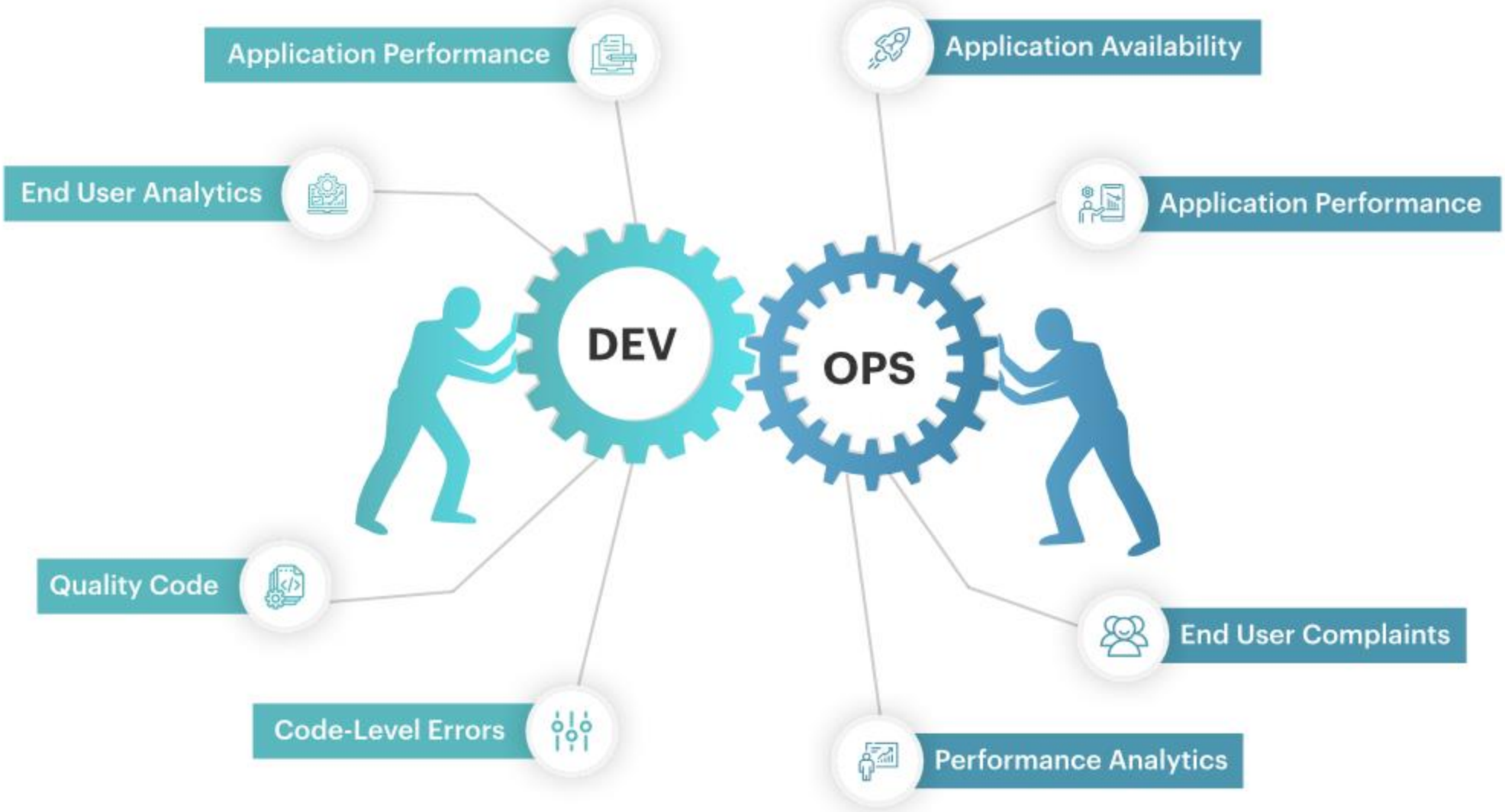


Image source: https://www.openxcell.com/devops

The DevOps Lifecycle

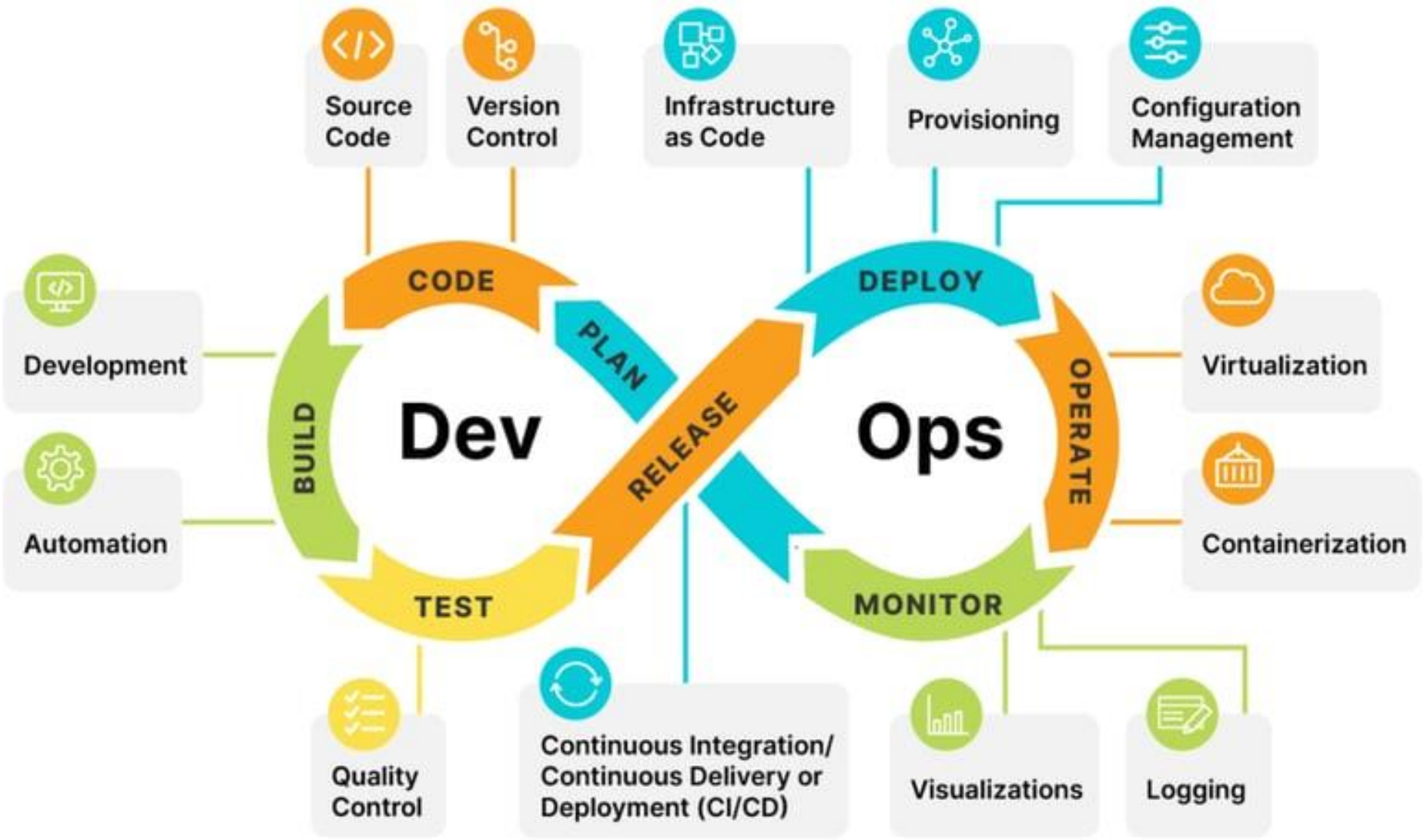
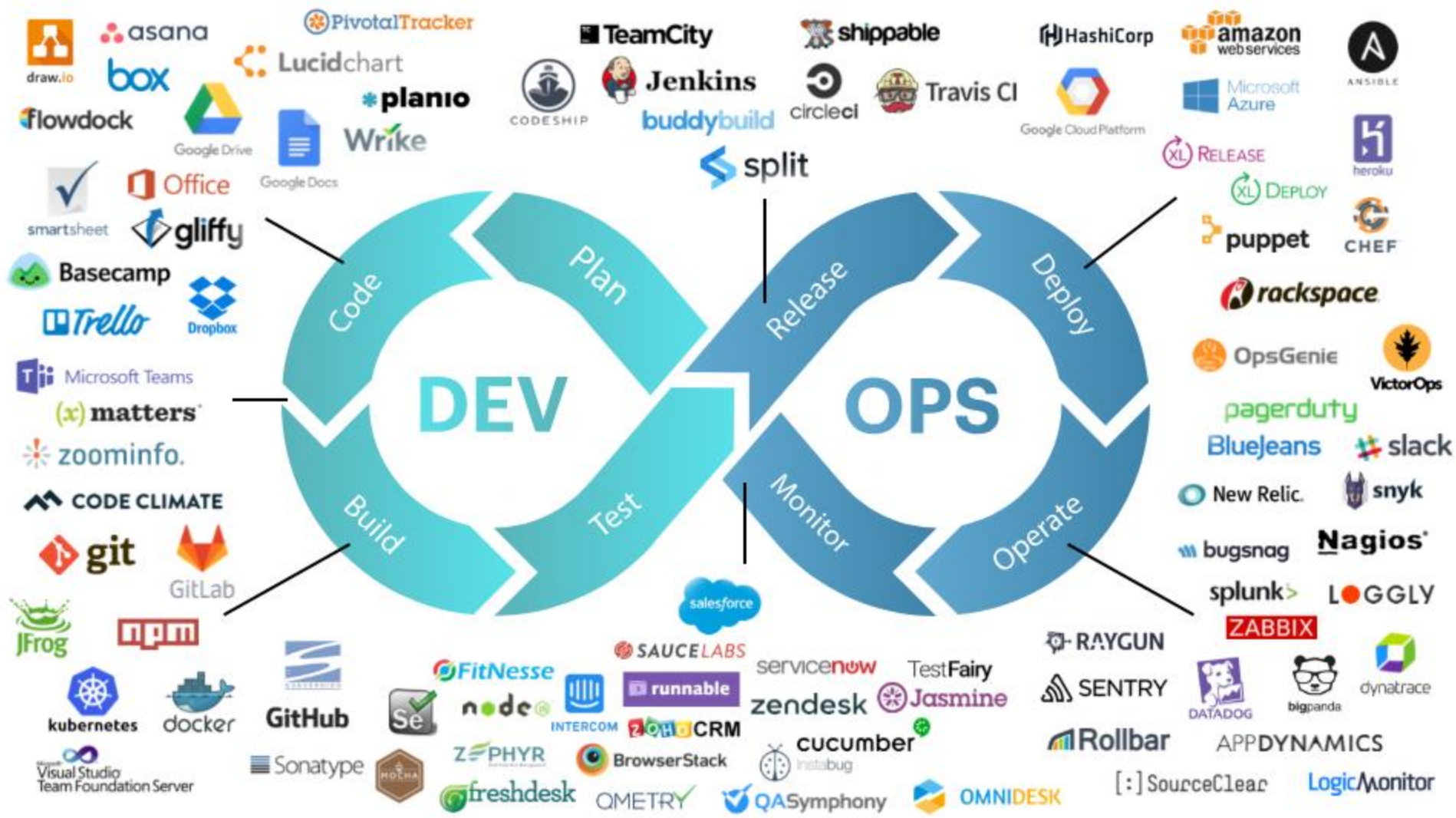


Image source: https://dev.to/rubixkube/the-future-of-devops-how-ai-is-shaping-infrastructure-management-3b95

DevOps Tools list according to DevOps Lifecycle



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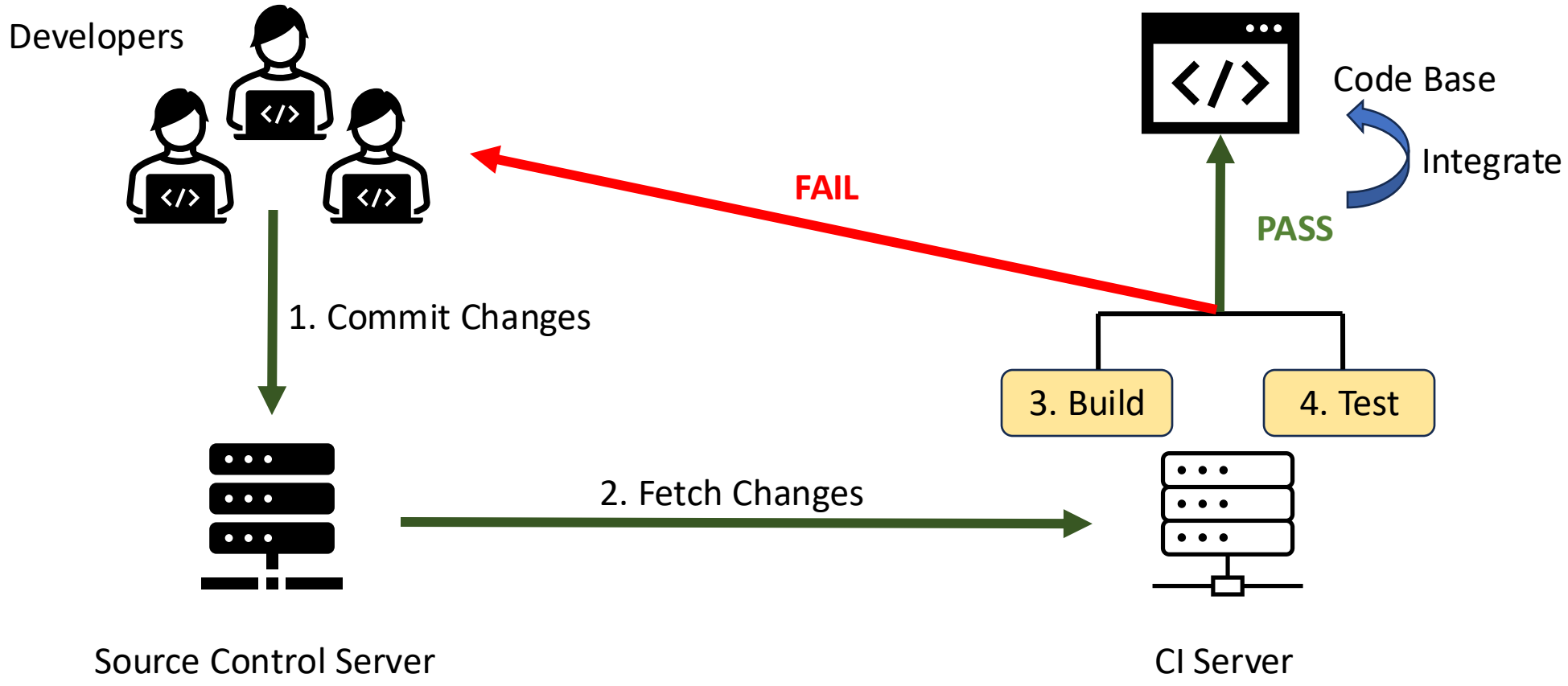
Continuous
Integration/Delivery/Deployment

Continuous Integration (CI)



Benefits of CI/CD

Faster Development Cycles:
Rapid integration and deployment lead to quicker releases.
Early Bug Detection:
Automated testing catches bugs early in the development process.
Consistent and Reliable Builds:
Ensures consistency across different environments.
Reduced Manual Intervention:
Automation minimizes the need for manual processes.

Continuous Integration/Deployment

Continuous Integration (CI):

- The practice of regularly merging code changes into a shared repository.
- Automated builds and tests to detect and fix integration issues early.

Continuous Deployment (CD):

- The automatic deployment of code changes to production or staging environments.
- Ensures a streamlined and efficient release process.

CI/CD Pipeline

Definition: A series of automated steps that code changes go through from development to deployment.

Stages:

- Code Compilation
- Automated Testing
- Artifact Generation
- Deployment to Staging
- User Acceptance Testing (UAT)
- Deployment to Production

Key Components of CI/CD

Version Control System (VCS):
e.g., Git, SVN - Enables collaborative development.

Build Automation Tools:
e.g., Jenkins, Circle CI - Automates building and packaging.

Automated Testing:
Unit tests, integration tests, and end-to-end tests.

Artifact Repository:
e.g., Nexus, Artifactory - Stores and manages build artifacts.

Continuous Integration

- **Purpose:** Ensure that code changes made by developers are integrated into the main codebase regularly and automatically.
- **Process:** Developers frequently merge their code changes into a shared repository, triggering an automated build and test process.
- **Benefits:** Early detection of integration problems, reduced integration risk, faster feedback on code changes, and increased collaboration among team members.

Continuous Deployment

- **Variation of Continuous Delivery:** Continuous Deployment is an even more automated extension of Continuous Delivery where every successful build that passes automated tests is automatically deployed to production without manual intervention.

- **Benefits:** Enables the most rapid and frequent release cycles, reducing time between development and delivery. **However, it requires a high level of confidence in automated testing and deployment processes**.

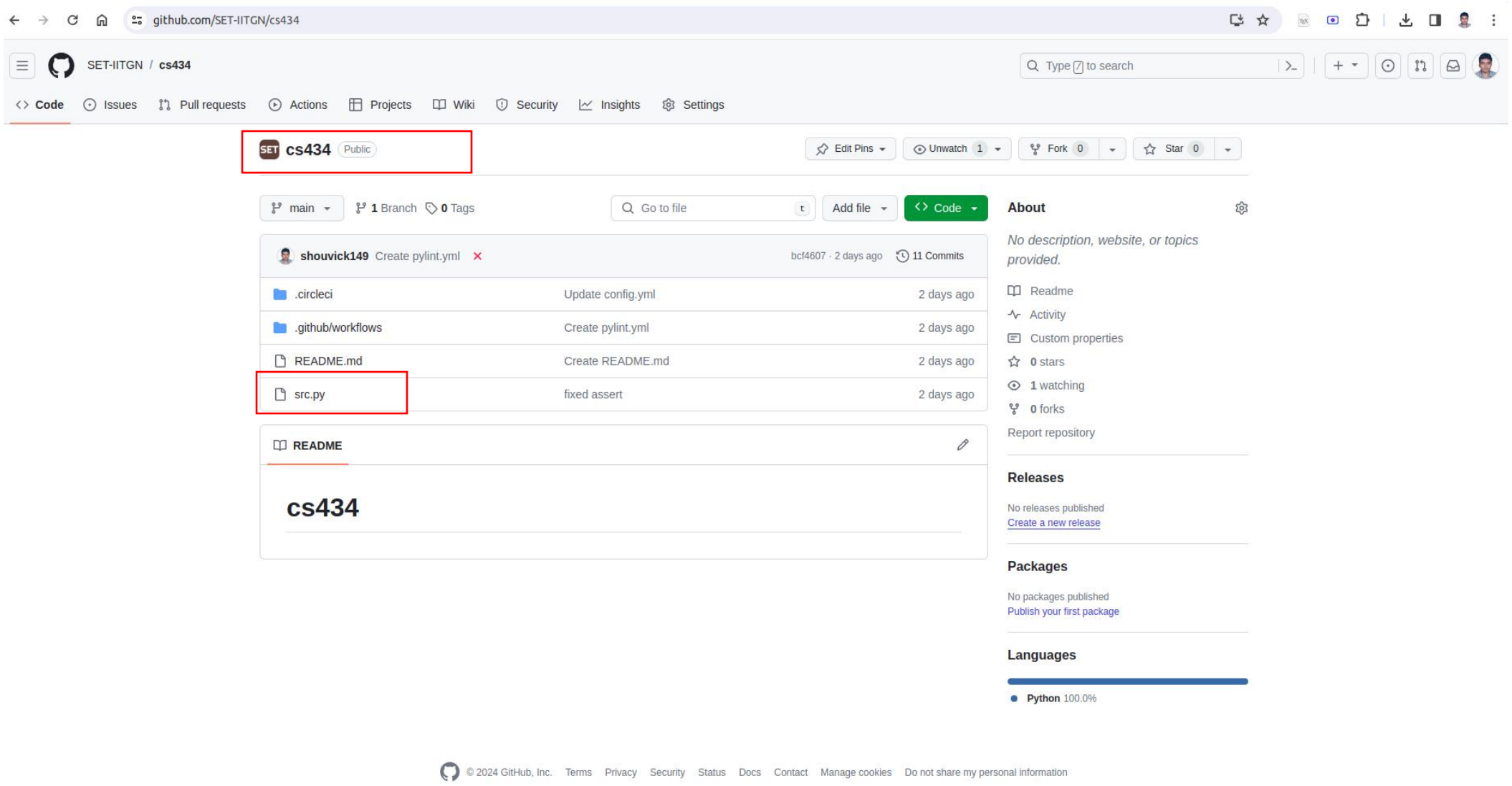
Continuous Delivery

- **Purpose:** Extend CI principles to ensure that the software can be released to production at any time by automating the entire software release process up to the point of deployment.

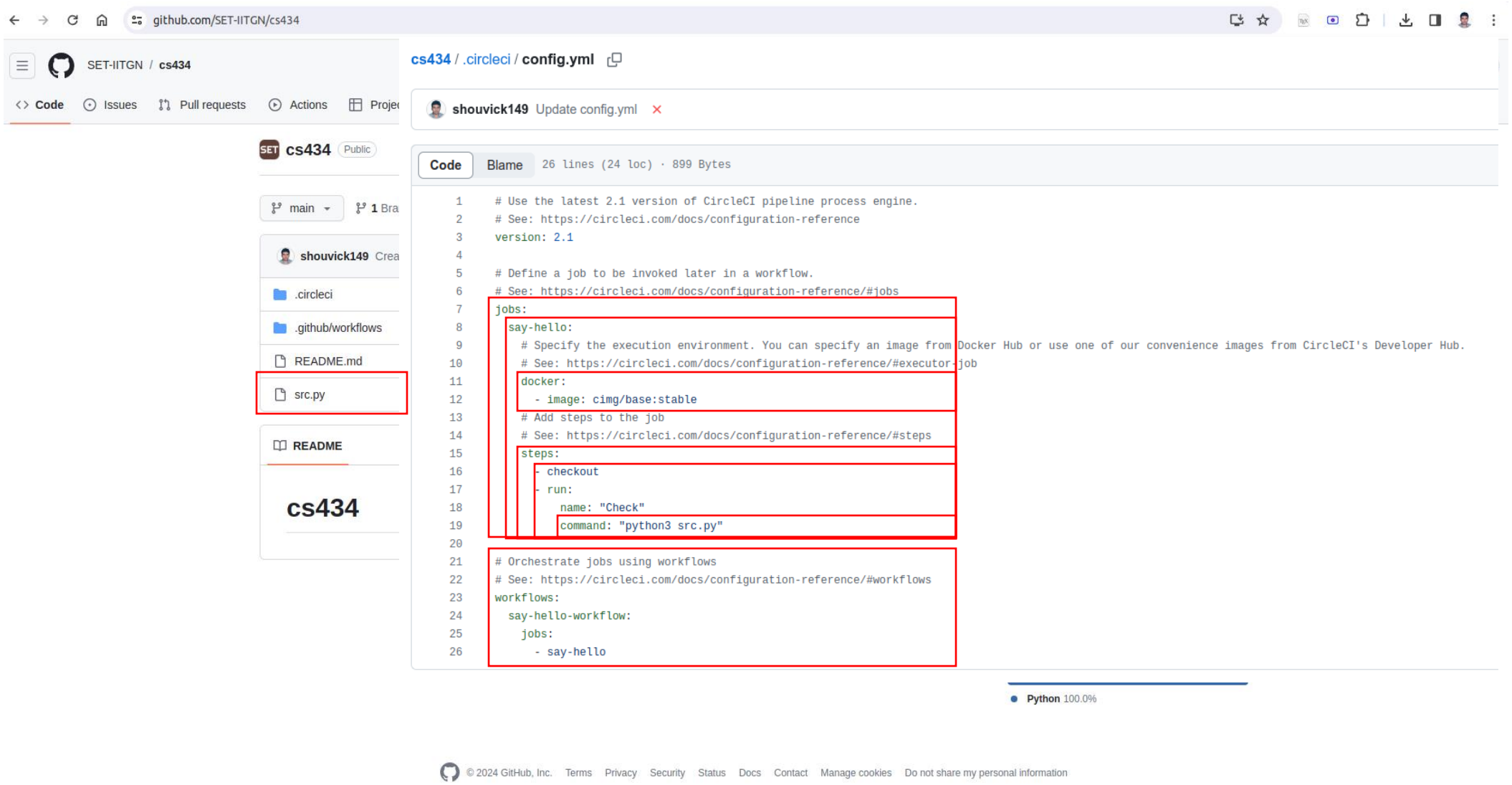
- **Process:** After successful CI, the software undergoes automated testing, and if all tests pass, it can be automatically deployed to a staging environment. It can then be manually or automatically promoted to production.

- **Benefits:** Reliable and repeatable software releases, faster time to market, reduced manual intervention in the release process, and the ability to release new features more frequently.

Setting up CI for a GitHub project (cs434)



Specify CI workflows/jobs in YAML (`config.yml`)



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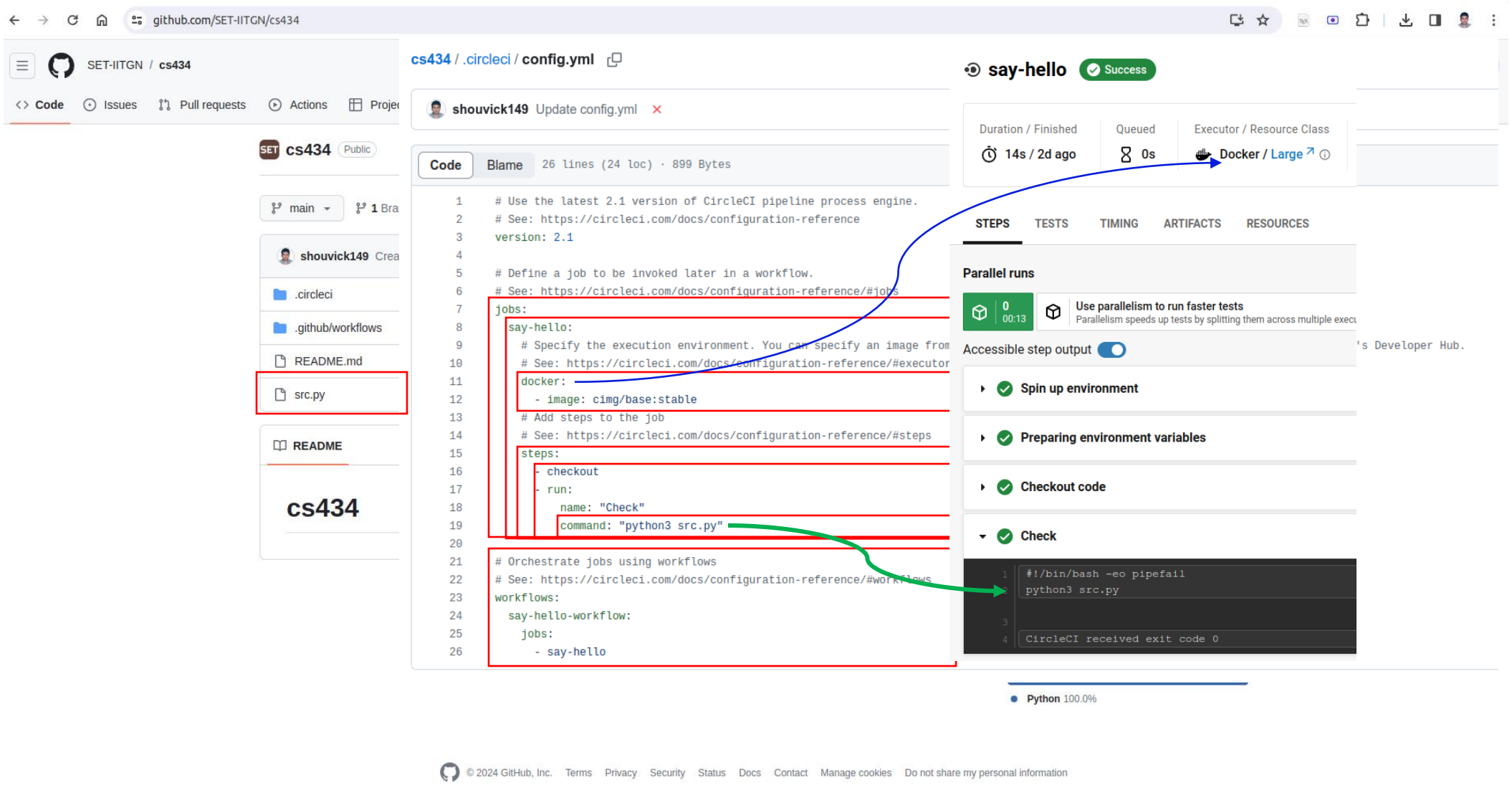

Texts, References, and Acknowledgements

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